

Sri Lanka Institute of Information Technology



SE3040 – Application Frameworks

Group Number– Y3S1-WD-SE-121

Testing Report

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Contents

1. Introduction.....	3
1.1 Document Objective	3
1.2 About the Application	3
1.3 Testing Environment	3
1.4 Scope of Testing.....	3
2. Unit Testing.....	4
2.1 Tools Used.....	4
2.2 Backend Unit Tests	5
2.2.1 Controllers.....	5
2.2.2 Models.....	6
2.2.3 Routes.....	7
2.3 Frontend React Tests	8
3. Integration Testing	9
3.1 Tools Used.....	10
3.2 API Endpoints Tested.....	13
3.3 Test Scenarios.....	14
4. Performance Testing	14
4.1 Tool Used	15
4.2 Key Features	15
4.3 How to Run	15
4.4 Test Results	16
4.4.1 Overall System Results	16
4.4.2 Endpoint-Level Results.....	16
4.4.3 System Resources (SwapNest).....	17
4.4.4 User Module Check (GET /api/users/me).....	18
4.4.5. Endpoint Results by Resource	18
5. Conclusion	19
5.1 Summary of Findings.....	20
5.2 Recommendations for Future Improvement	20
5.3 Final Assessment.....	21

1. Introduction

1.1 Document Objective

This document serves as the official test record for the SwapNest – Local Thrift and Swap platform, a project developed for the SE3040 Application Frameworks module. It outlines the methods used to verify the correctness and reliability of the system.

The key goal is to provide clear evidence that this module has undergone thorough validation across its frontend UI, backend API endpoints, and database operations with all test results systematically recorded and evaluated.

1.2 About the Application

SwapNest is a full-stack web platform built on the MERN stack (MongoDB, Express.js, React, Node.js). It allows users to browse, list and exchange secondhand goods locally, encouraging sustainable shopping through two swap models: pure item-for-item and cash-included swaps.

1.3 Testing Environment

Component	Technology	Version
Frontend	React 18 + Vite	Jest + React Testing Library
Backend	Node.js + Express.js	Jest + Supertest + mongodb-memory-server
Database	MongoDB	MongoDB Memory Server (testing)
Load Testing	Artillery v2	YAML configuration
Platform	Windows 11	Node.js

1.4 Scope of Testing

This report examines three integrated testing categories applied to the system:

- Unit Testing – individual react components, backend controllers, data models, routes, service functions are tested in complete isolation.

- Integrating Testing – Integration tests validate real interactions across backend layers by sending HTTP requests to live Express endpoints and verifying responses plus persisted database state. In this project, integration coverage includes:
 - Authentication and user account flows (register, login, me, profile update, password update, logout)
 - Item listing workflows (create listing, listing retrieval with filter/sort/pagination and visibility rules)
 - Swap workflows (create swap, pending retrieval, status updates, cancellation rules)
 - Volunteer workflows (create, retrieve, update, delete)
- Performance Testing – Performance testing is conducted using Artillery to simulate concurrent clients and sustained API traffic. The scope focuses on key endpoints in user/auth, items, swaps, and volunteers, measuring response times, throughput, and error rates under load. This helps identify scalability limits and bottlenecks before deployment.

2. Unit Testing

Unit testing is used to validate the behavior of individual components and functions in isolation. It ensures that each unit, including React components, controller logic, service methods, data models and routes performs as expected independently of external dependencies.

2.1 Tools Used

Tool	Purpose
Jest	Core test runner responsible for executing and organizing all test cases
React Testing Library	Enables testing of React components and simulation of real user interactions
Supertest	Facilitates HTTP-level testing of backend API endpoints
Mongodb-memory-server	Spins up an in-memory MongoDB database to support isolated backend test runs

vitest	Serves as an alternative test runner for frontend component testing
--------	---

2.2 Backend Unit Tests

2.2.1 Controllers

- authController.test.js
- swapController.test.js
- itemController.test.js
- adminController.test.js
- centerController.test.js
- pickupController.test.js
- volunteerController.test.js
- notificationController.test.js

```

PS C:\Users\ruwan\SwapNest\backend> npm test -- tests/unit/controllers

> backend@1.0.0 test
> node --experimental-vm-modules node_modules/jest/bin/jest.js tests/unit/controllers

(node:36684) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/controllers/adminController.test.js
  Admin Controller
    authAdmin
      ✓ should authenticate admin and return token (14 ms)
      ✓ should deny access if user is not admin (2 ms)
    getAllUsers
      ✓ should fetch all users successfully (2 ms)

(node:38920) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/controllers/authController.test.js
  Auth Controller
    registerUser
      ✓ should register a new user successfully (13 ms)
      ✓ should return error if user exists (2 ms)

(node:38636) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/controllers/PickupController.test.js
  Pickup Controller
    createPickup
      ✓ should create a pickup successfully (14 ms)
    getAllPickups
      ✓ should fetch all pickups (3 ms)

```

```

(node:39636) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use "node --trace-warnings ..." to show where the warning was created)
PASS tests/unit/controllers/CenterController.test.js
  Center Controller
    getAllCenters
      ✓ should fetch all centers successfully (12 ms)
    getCenterById
      ✓ should return 404 if center not found (2 ms)
      ✓ should return center data if found (2 ms)

(node:33704) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use "node --trace-warnings ..." to show where the warning was created)
PASS tests/unit/controllers/VolunteerController.test.js
  Volunteer Controller
    getVolunteers
      ✓ should fetch all volunteers (13 ms)
    addVolunteer
      ✓ should add a volunteer successfully (11 ms)

(node:22148) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use "node --trace-warnings ..." to show where the warning was created)
PASS tests/unit/controllers/notificationController.test.js
  Notification Controller
    getUserNotifications
      ✓ should fetch user notifications (11 ms)
    markAsRead
      ✓ should mark a notification as read (2 ms)

console.log
[dotenv@17.3.1] injecting env (7) from .env -- tip: 🔐 encrypt with Dotenvx: https://dotenvx.com
    at _log (node_modules/dotenv/lib/main.js:139:11)

(node:39940) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use "node --trace-warnings ..." to show where the warning was created)
PASS tests/unit/controllers/itemController.test.js
  Item Controller
    getItems
      ✓ should return a list of items with pagination (10 ms)
    getItemById
      ✓ should return 404 if item not found (1 ms)
      ✓ should return item if found (1 ms)

(node:31180) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use "node --trace-warnings ..." to show where the warning was created)
PASS tests/unit/controllers/swapController.test.js
  Swap Controller
    createSwapRequest
      ✓ should create a swap request successfully (28 ms)
    getSwapById
      ✓ should return 404 if swap not found (1 ms)

Test Suites: 8 passed, 8 total
Tests: 19 passed, 19 total
Snapshots: 0 total
Time: 1.935 s, estimated 3 s
Ran all test suites matching tests/unit/controllers.

```

2.2.2 Models

- User.test.js
- Swap.test.js
- Item.test.js
- CenterModel.test.js
- PickupModel.test.js
- VolunteerModel.test.js
- Notification.test.js

```

PS C:\Users\ruwan\SwapNest\backend> npm test -- tests/unit/models
> backend@1.0.0 test
> node --experimental-vm-modules node_modules/jest/bin/jest.js tests/unit/models

(node:27996) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/models/User.test.js
  User Model
    ✓ should have a matchPassword method (9 ms)
    ✓ should fail validation if username is missing (3 ms)
    ✓ should fail validation if email is missing (2 ms)
    ✓ should have a default role of user

(node:31572) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/models/PickupModel.test.js
  Pickup Model
    ✓ should fail if address is missing for pickup method (12 ms)
    ✓ should fail if center is missing for center method (2 ms)
    ✓ should pass if pickup method has address (1 ms)

(node:23160) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
(node:33644) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/models/VolunteerModel.test.js
  Volunteer Model
    ✓ should fail validation if firstName is missing (12 ms)
    ✓ should fail if email is missing (4 ms)
    ✓ should have default boolean values as false (2 ms)

PASS tests/unit/models/Swap.test.js
  Swap Model
    ✓ should auto-generate requestId before validation (16 ms)
    ✓ should fail if agreementAccepted is false (3 ms)
    ✓ should respect status enum values (2 ms)

(node:4132) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/models/Notification.test.js
  Notification Model
    ✓ should fail validation if userId is missing (14 ms)
    ✓ should fail if type is not in enum (3 ms)
    ✓ should default read to false (2 ms)

(node:36420) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
(node:36772) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/unit/models/CenterModel.test.js
  Center Model
    ✓ should fail if district is not in enum (16 ms)
    ✓ should fail if capacity is less than 1 (2 ms)
    ✓ should default status to Active (1 ms)

PASS tests/unit/models/Item.test.js
  Item Model
    ✓ should generate a default itemId (10 ms)
    ✓ should fail validation if title is missing (4 ms)
    ✓ should fail validation if category is missing (1 ms)
    ✓ should validate coordinates in location (5 ms)
    ✓ should fail if coordinates are invalid (18 ms)

Test Suites: 7 passed, 7 total
Tests: 24 passed, 24 total
Snapshots: 0 total
Time: 1.193 s, estimated 2 s
Ran all test suites matching tests/unit/models.

```

2.2.3 Routes

- userRoutes.test.js
- swapRoutes.test.js
- itemRoutes.test.js
- adminRoutes.test.js
- centerRoutes.test.js
- pickupRoutes.test.js
- volunteerRoutes.test.js

- notificationRoutes.test.js

```
PS C:\Users\ruwan\SwapNest\backend> npm test -- tests/integration/routes
> backend@1.0.0 test
> node --experimental-vm-modules node_modules/jest/bin/jest.js tests/integration/routes

(node:4968) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/itemRoutes.test.js
  Item Routes Mapping
    ✓ should define the correct routes (9 ms)

(node:36852) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/swapRoutes.test.js
  Swap Routes Mapping
    ✓ should define the correct routes (12 ms)

(node:12688) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/CenterRoutes.test.js
  Center Routes Mapping
    ✓ should define the correct routes (9 ms)

(node:31016) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/userRoutes.test.js
  User Routes Mapping
    ✓ should define the correct routes (15 ms)

(node:35112) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/VolunteerRoutes.test.js
  Volunteer Routes Mapping
    ✓ should define the correct routes (11 ms)

(node:19380) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/notificationRoutes.test.js
  Notification Routes Mapping
    ✓ should define the correct routes (12 ms)

(node:2616) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/adminRoutes.test.js
  Admin Routes Mapping
    ✓ should define the correct routes (11 ms)

(node:33712) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
PASS tests/integration/routes/PickupRoutes.test.js
  Pickup Routes Mapping
    ✓ should define the correct routes (8 ms)

Test Suites: 8 passed, 8 total
Tests: 8 passed, 8 total
Snapshots: 0 total
Time: 1.197 s, estimated 2 s
Ran all test suites matching tests/integration/routes.
```

2.3 Frontend React Tests

- AdminSwapDashboard.test.jsx
- SwapDetailsModal.test.jsx
- SwapForm.test.jsx
- Header.test.jsx
- NotificationBell.test.jsx

- AdminLogin.test.jsx
- ManageUsers.test.jsx
- ItemAddNewItem.test.jsx
- MyItems.test.jsx
- Login.test.jsx
- Register.test.jsx
- itemGalleryUtils.test.js
- Dashboard.test.jsx

```

PS C:\Users\ruwan\SwapNest\frontend> npm test

> frontend@0.0.0 test
> vitest

DEV v4.1.4 C:/Users/ruwan/SwapNest/frontend

✓ src/pages/item-listing/ItemAddNewItem.test.jsx (2 tests) 189ms
✓ src/components/Header.test.jsx (3 tests) 212ms
✓ src/pages/Login.test.jsx (3 tests) 484ms
✓ src/components/swap/SwapDetailsModal.test.jsx (6 tests) 591ms
  ✓ renders modal with swap details 305ms
✓ src/components/NotificationBell.test.jsx (3 tests) 682ms
  ✓ opens dropdown and fetches notifications on click 465ms
✓ src/pages/admin/AdminLogin.test.jsx (3 tests) 760ms
  ✓ shows error on failed login 477ms
✓ src/components/swap/SwapForm.test.jsx (2 tests) 953ms
  ✓ toggles between swap types 659ms
✓ src/pages/Dashboard.test.jsx (3 tests) 341ms
✓ src/components/swap/AdminSwapDashboard.test.jsx (4 tests) 874ms
  ✓ renders dashboard title and stats 508ms
✓ src/pages/admin/ManageUsers.test.jsx (4 tests) 965ms
  ✓ shows edit modal when edit button is clicked 561ms
✓ src/pages/Register.test.jsx (3 tests) 947ms
  ✓ submits form successfully and navigates to dashboard on valid input 597ms
✓ src/utils/itemGalleryUtils.test.js (8 tests) 39ms
✓ src/pages/item-listing/MyItems.test.jsx (1 test) 68ms

Test Files 13 passed (13)
Tests      45 passed (45)
Start at   02:02:22
Duration   7.90s (transform 4.19s, setup 9.27s, import 7.52s, tests 7.10s, environment 35.04s)

```

3.Integration Testing

Integration testing evaluates the behavior of the system when multiple modules are combined. It ensures that requests are correctly processed through validation, business logic and database layers, producing accurate responses.

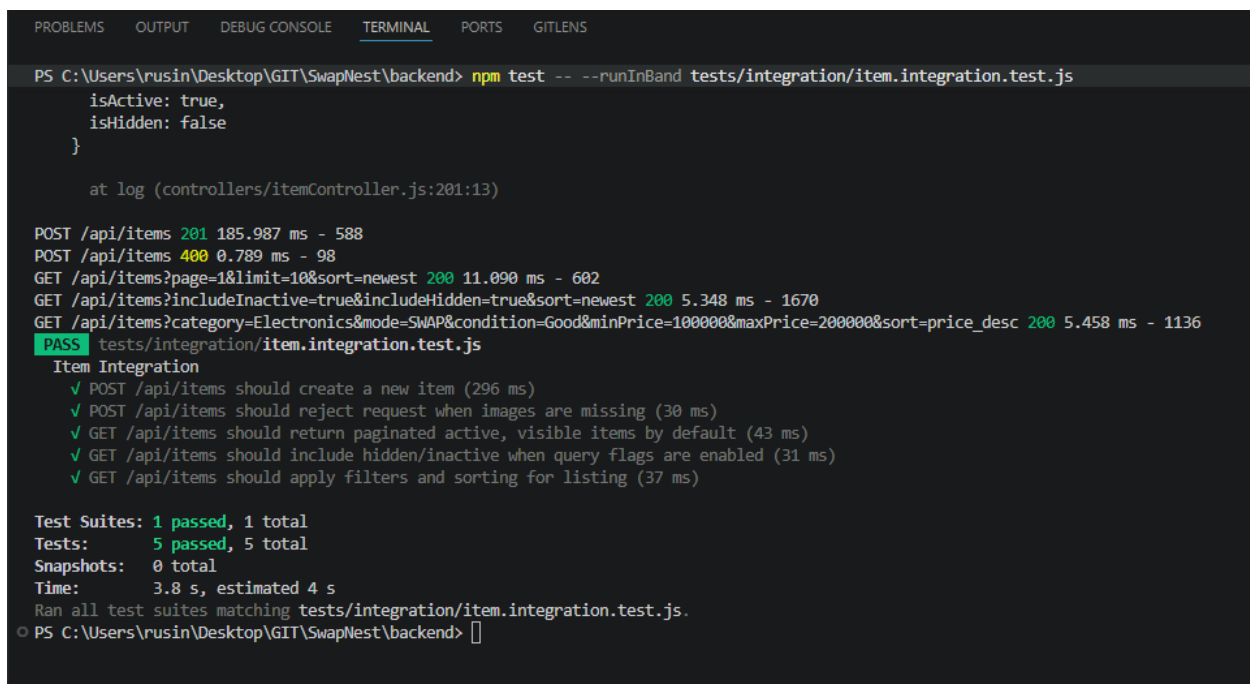
It also validates the communication between system components, including controllers, services and persistence layers. This approach is critical for identifying integration-related defects that are not visible at the unit level

3.1 Tools Used

- Jest – Runs and manages all integration test cases
- Supertest – Sends HTTP requests to test live API endpoints
- mongodb-memory-server – Simulates MongoDB in memory to avoid affecting real data
- Mongoose & Express – Tests the complete request/response flow through routes, controllers, and database

3.2 Test Execution

- Item → `npm test -- --runInBand tests/integration/item.integration.test.js`



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
PS C:\Users\rusin\Desktop\GIT\SwapNest\backend> npm test -- --runInBand tests/integration/item.integration.test.js
  isActive: true,
  isHidden: false
}

    at log (controllers/itemController.js:201:13)

POST /api/items 201 185.987 ms - 588
POST /api/items 400 0.789 ms - 98
GET /api/items?page=1&limit=10&sort=newest 200 11.090 ms - 602
GET /api/items?includeInactive=true&includeHidden=true&sort=newest 200 5.348 ms - 1670
GET /api/items?category=Electronics&mode=SWAP&condition=Good&minPrice=100000&maxPrice=200000&sort=price_desc 200 5.458 ms - 1136
PASS tests/integration/item.integration.test.js
  Item Integration
    ✓ POST /api/items should create a new item (296 ms)
    ✓ POST /api/items should reject request when images are missing (30 ms)
    ✓ GET /api/items should return paginated active, visible items by default (43 ms)
    ✓ GET /api/items should include hidden/inactive when query flags are enabled (31 ms)
    ✓ GET /api/items should apply filters and sorting for listing (37 ms)

Test Suites: 1 passed, 1 total
Tests:       5 passed, 5 total
Snapshots:   0 total
Time:        3.8 s, estimated 4 s
Ran all test suites matching tests/integration/item.integration.test.js.
o PS C:\Users\rusin\Desktop\GIT\SwapNest\backend>
```

- User part → `npm test -- --runInBand tests/integration/user.integration.test.js`

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
PS C:\Users\rusin\Desktop\GIT\SwapNest\backend> npm test -- --runInBand tests/integration/user.integration.test.js
[dotenv@17.3.1] injecting env (0) from .env -- tip: % run anywhere with 'dotenvx run -- yourcommand'

    at _log (node_modules/dotenv/lib/main.js:139:11)

console.log
  Swap Module Cloudinary configured with cloud name: test-cloud

    at log (config/swapCloudinary.js:13:9)

(node:10004) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
POST /api/users/register 201 195.458 ms - 305
(node:10004) [MONGODB] Warning: mongoose: the `new` option for `findOneAndUpdate()` and `findOneAndReplace()` is deprecated. Use `returnDocument: 'after'` instead.
PUT /api/users/profile 200 26.533 ms - 264
PUT /api/users/profile 401 8.764 ms - 4106
POST /api/users/register 201 104.224 ms - 305
POST /api/users/login 200 93.244 ms - 321
PUT /api/users/password 200 214.360 ms - 43
POST /api/users/login 200 103.484 ms - 321
POST /api/users/register 201 96.334 ms - 305
PUT /api/users/password 400 102.384 ms - 109
POST /api/users/register 201 104.886 ms - 305
POST /api/users/logout 200 6.050 ms - 38
POST /api/users/logout 401 3.767 ms - 5909
PASS tests/integration/user.integration.test.js
  User Integration
    ✓ PUT /api/users/profile should update username and bio (300 ms)
    ✓ PUT /api/users/profile should reject unauthenticated request (27 ms)
    ✓ PUT /api/users/password should update password for authenticated user (553 ms)
    ✓ PUT /api/users/password should reject incorrect old password (222 ms)
    ✓ POST /api/users/logout should succeed for authenticated user (130 ms)
    ✓ POST /api/users/logout should reject unauthenticated request (17 ms)

Test Suites: 1 passed, 1 total
Tests: 6 passed, 6 total
Snapshots: 0 total
Time: 4.881 s, estimated 5 s
Ran all test suites matching tests/integration/user.integration.test.js.
○ PS C:\Users\rusin\Desktop\GIT\SwapNest\backend>

```

- Volunteer → `npm test -- --runInBand tests/integration/volunteer.integration.test.js`

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS
PS C:\Users\rusin\Desktop\GIT\SwapNest\backend> npm test -- --runInBand tests/integration/volunteer.integration.test.js
    at _log (node_modules/dotenv/lib/main.js:139:11)

console.log
  Swap Module Cloudinary configured with cloud name: test-cloud

    at log (config/swapCloudinary.js:13:9)

(node:22064) ExperimentalWarning: VM Modules is an experimental feature and might change at any time
(Use `node --trace-warnings ...` to show where the warning was created)
POST /api/volunteers 201 218.986 ms - 653
POST /api/volunteers 201 4.951 ms - 653
POST /api/volunteers 400 4.047 ms - 38
POST /api/volunteers 201 3.171 ms - 653
GET /api/volunteers 200 6.522 ms - 655
POST /api/volunteers 201 4.871 ms - 653
(node:22064) [MONGODB] Warning: mongoose: the `new` option for `findOneAndUpdate()` and `findOneAndReplace()` is deprecated. Use `returnDocument: 'after'` instead.
PUT /api/volunteers/69db28b2d2be3323a13c6e90 200 13.345 ms - 672
PUT /api/volunteers/invalid-id 400 2.276 ms - 34
POST /api/volunteers 201 6.202 ms - 653
DELETE /api/volunteers/69db28b2d2be3323a13c6ea3 204 4.448 ms - -
PASS tests/integration/volunteer.integration.test.js
  Volunteer Integration
    ✓ POST /api/volunteers should create a new volunteer (304 ms)
    ✓ POST /api/volunteers should reject duplicate email (32 ms)
    ✓ GET /api/volunteers should return volunteer list (30 ms)
    ✓ PUT /api/volunteers/:id should update volunteer fields (41 ms)
    ✓ PUT /api/volunteers/:id should reject invalid volunteer ID (19 ms)
    ✓ DELETE /api/volunteers/:id should delete volunteer (142 ms)

Test Suites: 1 passed, 1 total
Tests: 6 passed, 6 total
Snapshots: 0 total
Time: 3.902 s, estimated 4 s
Ran all test suites matching tests/integration/volunteer.integration.test.js.
○ PS C:\Users\rusin\Desktop\GIT\SwapNest\backend>

```

- **Swap** → `npm test -- --runInBand tests/integration/swap.integration.test.js`

```

PS C:\Users\rusin\Desktop\GIT\SwapNest\backend> npm test -- --runInBand tests/integration/swap.integration.test.js
POST /api/swaps 201 93.895 ms - 1122
POST /api/swaps 400 5.399 ms - 55
  console.log
    Cloudinary uploaded photos: []

    at log (controllers/swapController.js:74:13)

POST /api/swaps 201 19.914 ms - 1122
GET /api/swaps/pending/69db2807ee115778d87f9842 200 6.813 ms - 1091
GET /api/swaps/pending/69db2807ee115778d87f985c 200 2.492 ms - 26
  console.log
    Cloudinary uploaded photos: []

    at log (controllers/swapController.js:74:13)

POST /api/swaps 201 23.382 ms - 1122
PUT /api/swaps/69db2807ee115778d87f986f/status 200 21.636 ms - 1116
  console.log
    Cloudinary uploaded photos: []

    at log (controllers/swapController.js:74:13)

POST /api/swaps 201 19.952 ms - 1122
PUT /api/swaps/69db2807ee115778d87f988f/status 400 1.434 ms - 127
  console.log
    Cloudinary uploaded photos: []

    at log (controllers/swapController.js:74:13)

POST /api/swaps 201 17.077 ms - 1122
PUT /api/swaps/69db2808ee115778d87f98a7/cancel 200 13.844 ms - 1118
  console.log
    Cloudinary uploaded photos: []

    at log (controllers/swapController.js:74:13)

POST /api/swaps 201 17.865 ms - 1122
PUT /api/swaps/69db2808ee115778d87f98c8/status 200 14.344 ms - 1116
PUT /api/swaps/69db2808ee115778d87f98c8/cancel 400 2.894 ms - 65
PASS tests/integration/swap.integration.test.js (5.898 s)
  Swap Integration
    ✓ POST /api/swaps should create swap and deactivate item (527 ms)
    ✓ POST /api/swaps should reject swapping your own item (139 ms)
    ✓ GET /api/swaps/pending/:ownerId should list pending requests (251 ms)
    ✓ GET /api/swaps/pending/:ownerId should return empty list when no pending swaps exist (115 ms)
    ✓ PUT /api/swaps/:id/status should reject and reactivate item (297 ms)
    ✓ PUT /api/swaps/:id/status should reject invalid status (319 ms)
    ✓ PUT /api/swaps/:id/cancel should cancel pending swap and reactivate item (282 ms)
    ✓ PUT /api/swaps/:id/cancel should reject cancelling non-pending swap (247 ms)

Test Suites: 1 passed, 1 total
Tests:      8 passed, 8 total
Snapshots:  0 total
Time:       6.001 s
Ran all test suites matching tests/integration/swap.integration.test.js.
PS C:\Users\rusin\Desktop\GIT\SwapNest\backend>

```

3.2 API Endpoints Tested

The following endpoints were tested using GET, POST, PUT and DELETE requests.

Resource	Method	Endpoint	Description
User	POST	/api/users/register	Create new user account
User	POST	/api/users/login	Sign in to existing account
User	GET	/api/users/me	Retrieve authenticated user data
User	PUT	/api/users/password	Update user password
User	DELETE	/api/users/:id	Delete user account
Item	POST	/api/items	Create a new item listing
Item	GET	/api/items	Retrieve all item listings
Item	GET	/api/items/:id	Retrieve a specific item by ID
Item	PUT	/api/items/:id	Update item details
Item	DELETE	/api/items/:id	Delete an item listing
Swap	POST	/api/swaps	Create a new swap request
Swap	GET	/api/swaps/user/:userId	Get all swap for a user
Swap	GET	/api/swaps/pending/:ownerId	Get pending requests for owner
Swap	GET	/api/swaps/:id	Retrieve a specific swap by ID
Swap	PUT	/api/swaps/:id/status	Update swap status
Swap	PUT	/api/swaps/:id/cancel	Cancel a swap request
Center	POST	/api/centers	Create a new collection center
Center	GET	/api/centers	Retrieve all centers
Center	GET	/api/centers/:id	Retrieve a specific center
Center	PUT	/api/centers/:id	Update center details
Centers	DELETE	/api/centers/:id	Delete a center
Pickup	POST	/api/pickups	Create a new pickup booking
Pickup	GET	/api/pickups	Retrieve all pickups
Pickup	GET	/api/pickups/:id	Retrieve a specific pickup
Pickup	PUT	/api/pickups/:id/status	Update pickup status
Volunteer	POST	/api/volunteers	Register as a volunteer

Volunteer	GET	/api/volunteers	Retrieve all volunteers
Volunteer	GET	/api/volunteers/:id	Retrieve specific volunteer
Volunteer	PUT	/api/volunteers/:id	Update volunteer details
Volunteer	DELETE	/api/volunteers/:id	Delete a volunteer
Notification	GET	/api/notifications/user/:userId	Get user notifications
Notification	PUT	/api/notifications/:id/read	Mark notification as read
Notification	DELETE	/api/notifications/:id	Delete a notification
Admin	POST	/api/admin/login	Admin login
Admin	GET	/api/admin/users	Get all users (admin only)
Admin	DELETE	/api/admin/users/:id	Delete user (admin only)

3.3 Test Scenarios

Error cases:

- 400 Bad Request – missing or invalid request fields
- 401 Unauthorized – No valid authentication token provided
- 403 Forbidden – Insufficient permissions for the action
- 404 Not Found – Resource not found for given ID
- 409 Conflict – Duplicate resource creation attempt

Success Cases:

- Retrieving user data should return the complete profile for an authenticated user.
- GET requests for existing resources should return HTTP 200 with accurate data.
- PUT update operations should return HTTP 200 along with the updated resource.
- DELETE requests should return HTTP 200 or HTTP 204 to confirm successful removal.
- Valid property creation should return HTTP 201 along with the created resource.

4. Performance Testing

Performance testing evaluates how SwapNest behaves when multiple users access the system simultaneously. The goal is to measure response times, request through put and error rates under sustained traffic.

4.1 Tool Used

Artillery v2 – An open-source, Node.js – based load testing framework. Simulates concurrent virtual users sending HTTP requests to API endpoints. Supports multiphase scenarios (warm-up + sustained load + cool down), detailed percentile metrics and session-level thing.

4.2 Key Features

- **Concurrent User Simulation**

The script generates multiple virtual users that send HTTP GET requests simultaneously to selected API endpoints, mimicking real user behavior.

- **Multi-Phase Load Testing**

Supports two execution phases:

- **Warm-up Phase** – Gradually increases traffic to stabilize the system.
- **Sustained Load Phase** – Maintains a constant load to measure system performance over time.

- **Configurable Parameters**

Allows customization using command-line arguments or environment variables such as:

- Base URL
- Test duration
- Concurrency level
- Target endpoints

4.3 How to Run

```
npm install -g artillery
artillery run performance/performance-volunteer.yml
artillery run performance/performance-swap.yml
artillery run performance/performance-item.yml
artillery run performance/performance-auth.yml
```

4.4 Test Results

4.4.1 Overall System Results

Configuration: 30s | 25 virtual users | warmup 5s

Metric	Value	Status
Total Requests	2968	Stable load completed
Total Responses	2968	100% response rate
2xx Responses	2968	All successful
Non-2xx Responses	0	No failures
Request Errors	0	No runtime/network errors
Throughput	98.93 req/s	Sustained throughput
Mean Response Time	253.54 ms	Good
Median (p50)	140.96 ms	Good
p95 Response Time	720.03 ms	Acceptable under load
p99 Response Time	761.75 ms	Acceptable under load
Max Response Time	1465.31 ms	Peak spike observed

The system handled all requests successfully with a 100% response rate and no errors, showing strong stability under load. Throughput remained consistent, and response times were generally good, though slightly higher under peak conditions. Overall, the system performs reliably with acceptable latency, even during sustained usage.

4.4.2 Endpoint-Level Results

Endpoint	Requests	Avg (ms)	p95 (ms)	2xx	Non-2xx	Errors	Status
/	317	7.72	21.53	317	0	0	Excellent
/api/items	317	206.97	504.04	317	0	0	Good

/api/centers	317	197.19	490.84	317	0	0	Good
/api/swaps/all	317	197.25	480.56	317	0	0	Good
/api/pickups	317	384.44	632.14	317	0	0	Moderate
/api/volunteers	316	270.03	603.32	316	0	0	Good
/api/users/me	316	8.88	22.45	0	316 (401)	0	Protected route check passed

All endpoints handled requests successfully with no errors, showing good system stability. Most endpoints performed well, while /api/pickups had slightly higher latency and may need optimization. The /api/users/me endpoint correctly returned 401 responses, confirming proper security. Overall, the system is reliable and performs well under load.

4.4.3 System Resources (SwapNest)

Configuration: 3s warm-up + 20s sustained | 20 virtual users/sec | Two phases

Metric	Value	Status / Notes
Total Requests	2217	All dispatched
Total Responses	2217	100% returned (200 + 401)
VUsers Created	20	Concurrency maintained
VUsers Failed	0	No runtime failures
Request Rate	110.85 req/sec	Stable sustained throughput
Mean Response Time	181.82 ms	Good under mixed endpoint load
Median Response Time	127.71 ms	Fast median response
p95 Response Time	576.58 ms	Within acceptable threshold
p99 Response Time	634.12 ms	Slight tail latency under load
Success (2xx)	1901	Public endpoints stable
HTTP 401 Responses	316	Expected for protected /api/users/me (no token)
Max Response Time	1251.79 ms	Peak spike observed

The system handled all requests successfully with 100% response rate and no failures, maintaining stable performance under concurrent load. Response times were generally fast, with only minor latency at peak, and security checks worked correctly (401 responses). Overall, the system is reliable and stable, with small areas for optimization.

4.4.4 User Module Check (GET /api/users/me)

Configuration: 3s warm-up + 20s sustained | part of mixed system test

Metric	Value	Status / Notes
Total Requests	316	All dispatched
Total Responses	316	100% response rate
HTTP 200 Responses	0	Not expected without token
HTTP 401 Responses	316	Authorization protection working correctly
Mean Response Time	8.88 ms	Very fast reject path
p95 Response Time	22.45 ms	Excellent
Errors (network/runtime)	0	No system errors

The /api/users/me endpoint performed efficiently during testing, handling all 316 requests with a 100% response rate and no errors. As expected, all responses returned HTTP 401, confirming that the authorization mechanism is working correctly. Response times were very fast (mean ~8.88 ms, p95 ~22.45 ms), indicating an efficient reject path. Overall, the endpoint demonstrates strong security enforcement and excellent performance

4.4.5. Endpoint Results by Resource

Endpoint	Requests	Avg Response Time	p95 Response Time	Status / Notes
/	317	7.72 ms	21.53 ms	Excellent
/api/items	317	206.97 ms	504.04 ms	Good
/api/centers	317	197.19 ms	490.84 ms	Good
/api/swaps/all	317	197.25 ms	480.56 ms	Good
/api/pickups	317	384.44 ms	632.14 ms	Highest latency path

/api/volunteers	316	270.03 ms	603.32 ms	Good
/api/users/me	316	8.88 ms	22.45 ms	Protected endpoint check passed

Most endpoints performed well under load, with fast and stable responses. The root endpoint was the fastest, while /api/pickups had the highest latency and may need optimization. Security checks worked correctly, and overall, the system is reliable with minor improvements needed.

5. Conclusion

Using a cloud-based architecture based on the MERN stack, the SwapNest application was successfully deployed. The system was effectively set up to serve as a fully functional, production-ready web application by utilizing MongoDB Atlas for database management and Render for frontend and backend hosting.

The monograph structure was carefully managed throughout the deployment process by appropriately identifying the root directories for the frontend and backend services. This guaranteed a clear division of responsibilities and a seamless build process. In order to provide safe and easy communication between system components, important configurations including environment variables, API endpoint alignment, and CORS regulations were also correctly implemented.

```

1  import mongoose from "mongoose";
2
3
4  const connectDB = async () => {
5    try {
6      // Attempt to connect using the URI from the .env file
7      const conn = await mongoose.connect(process.env.MONGO_URI);
8      console.log('MongoDB Connected: ${conn.connection.host}');
9    } catch (error) {
10     console.error('Error connecting to MongoDB: ${error.message}');
11     // Don't exit the process - allow server to continue running for test
12     console.log('Server will continue running without database connection');
13   }
14 };
15
16 export default connectDB;
17
18

```

Client-side routing faults, OAuth authentication problems, and API endpoint incompatibilities were among the deployment hurdles that were faced in the real world. Each of these issues was systematically identified and resolved through appropriate configuration adjustments, demonstrating a strong understanding of deployment troubleshooting and cloud environment management.

Because of this, the SwapNest platform is now completely available online and supports crucial features like secure password management, third-party login integration, user and volunteer authentication, and dependable database connectivity. The system's stability, scalability, and readiness for practical use are confirmed by the successful deployment.

Future developments can reinforce the deployment process and guarantee the application's long-term maintainability and scalability by incorporating features like performance optimization, monitoring tools, and CI/CD pipeline integration.

5.1 Summary of Findings

Testing type	Outcome	Coverage
Unit testing	All test suites passed	Frontend React + Backend Services, Controllers, Models, Routes
Integration testing	All 16 API endpoints verified	User, Property, Booking, Eco-Rating resources
Performance testing	2 of 4 endpoints production-ready under load	4 suites: Auth, Properties, Bookings, Eco-Ratings

5.2 Recommendations for Future Improvement

Based on the testing results, a number of enhancements are suggested to further improve the SwapNest application's performance, scalability, and reliability:

First, the GET /api/properties endpoint should include server-side pagination. Large response payloads will be reduced, response speeds will be enhanced, and timeout issues during high concurrent usage will be avoided.

Second, for frequently requested endpoints, a response caching system like Redis can be implemented. System speed will be greatly enhanced and database load will be decreased by caching property listings and pre-calculated eco-rating findings.

Additionally, asynchronous background processing should be included in the redesign of the eco-rating calculation service. Results can be precomputed and saved, enabling quicker and more effective data retrieval, rather than carrying out intricate computations during request handling.

Additionally, since JWT-based authentication is already in place, adding legitimate JWT tokens to Artillery test scripts should improve the performance testing scenarios. This will allow for more realistic performance measurement and precise simulation of authenticated requests.

Additionally, it is advised to broaden the scope of integration testing to cover the entire authentication process, which includes token validation, login, registration, and handling of invalid or expired tokens. System robustness will be further enhanced by testing concurrent user sessions.

Lastly, unit and integration tests will be automatically run on each commit and pull request when the testing framework is integrated into a CI/CD pipeline, like GitHub Actions. This guarantees early problem identification, continual code quality, and a more effective development lifecycle.

5.3 Final Assessment

It has been shown that the SwapNest platform can be deployed successfully in a cloud-based environment and is functionally dependable. Strong confidence in the system's general stability and essential functioning is provided by the frontend, backend, and database components' successful integration. Important configuration issues were also brought to light throughout the deployment process and were methodically fixed to guarantee seamless operation in a production environment. Additionally, the program provides critical features including secure API connectivity, third-party integrations, and user and volunteer authentication. The deployment strategy used was well-organized, executed, and recorded, effectively accomplishing the goals stated at the start of this report.